

**CRANBROOK**

**MATHEMATICS EXTENSION 2**

**2008**

**HIGHER SCHOOL CERTIFICATE**

**TRIAL EXAMINATION**

**General Instructions**

- Reading time – 5 minutes
- Writing time – 3 hours
- All eight questions should be attempted
- Total marks available - 120
- All questions are worth 15 marks
- An approved calculator may be used
- All relevant working should be shown for each question
- Answer each question in a separate 8 page booklet.
- Standard integrals sheet at back of examination.

**Question 1** (15 marks)

Marked by SKB

**Marks**

(a) Find  $\int x \tan x^2 dx$ .

**2**

(b) Use the substitution  $u = \sqrt{x}$  to evaluate  $\int_4^9 \frac{x}{\sqrt{x}(1+x)} dx$ .

**3**

(c) Use the completion of squares method to find  $\int \frac{-2}{\sqrt{3+2x-x^2}} dx$ .

**2**

(d) (i) Find the real numbers  $a$ ,  $b$  and  $c$  such that

**2**

$$\frac{2x^2 + 2x + 5}{(x^2 + 2)(1-x)} = \frac{ax + b}{x^2 + 2} + \frac{c}{1-x}.$$

(ii) Hence find  $\int \frac{2x^2 + 2x + 5}{(x^2 + 2)(1-x)} dx$ .

**2**

(e) Use integration by parts to evaluate  $\int_1^5 \frac{\ln x}{\sqrt{x}} dx$ .

**4**

| Question 2 (15 marks)                                                                                                      | Marked by SKB | Marks |
|----------------------------------------------------------------------------------------------------------------------------|---------------|-------|
| (a) Evaluate $\int_{-1}^1 \frac{\tan^{-1} x}{1+x^4} dx$                                                                    |               | 2     |
| (b) Evaluate $\int_0^1 \sqrt{4-x^2} dx$                                                                                    |               | 4     |
| (c) By using the result $\int_0^a f(x) dx = \int_0^a f(a-x) dx$<br>evaluate $\int_0^{2\pi} \frac{x \cos x}{1+\sin^2 x} dx$ |               | 3     |
| (d) Let $I_n = \int_0^1 x(x^2-1)^n dx$ for $n = 0, 1, 2, \dots$                                                            |               |       |
| (i) Use integration by parts to show that                                                                                  |               | 3     |
| $I_n = \frac{-n}{n+1} I_{n-1} \text{ for } n \geq 1.$                                                                      |               |       |
| (ii) Hence or otherwise show that                                                                                          |               | 2     |
| $I_n = \frac{(-1)^n}{2(n+1)} \text{ for } n \geq 0.$                                                                       |               |       |
| (iii) Explain why $I_{2n} > I_{2n+1}$ for $n \geq 0$                                                                       |               | 1     |

## Question 3 (15 marks)

Marked by JSH

Marks

- (a) Let  $z = 3 - i$  and  $w = 2 + 4i$ .  
Find the following in the form  $x + yi$ .

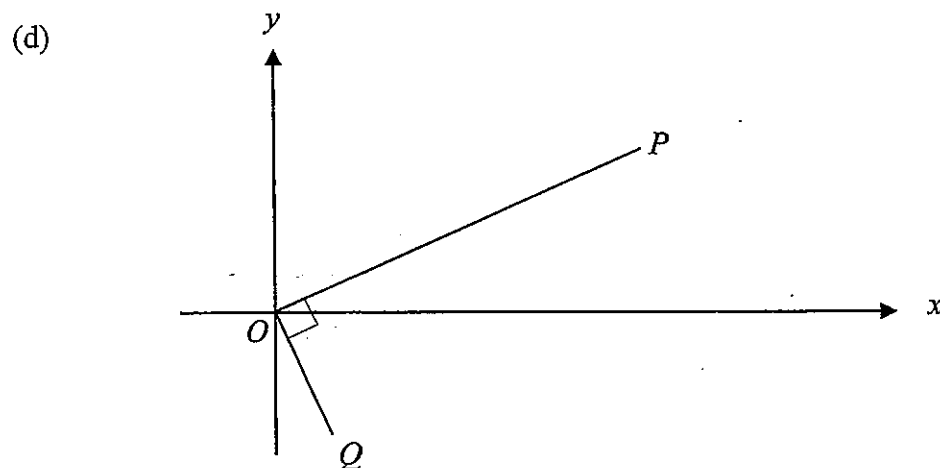
(i)  $z\bar{w}$  1

(ii)  $\frac{z}{w}$  1

- (b) (i) Express  $1 + i$  in modulus-argument form. 2

- (ii) Hence, find the values of  $n$ , for which 3  
 $(1 + i)^n + (1 - i)^n = 0$   
 where  $n$  is a positive integer.

- (c) Sketch the region in the Argand diagram where the inequalities 2  
 $|z - 1| \leq 1$  and  $\frac{\pi}{4} \leq \arg(z - 1) \leq \frac{\pi}{2}$  both hold.



In the Argand diagram above, point  $P$  corresponds to the complex number  $z$ . 1

The triangle  $OPQ$  is a right-angled triangle and  $OP = 3OQ$ .  
 What is the complex number that corresponds to point  $Q$ ?

- (e) (i) Find all the solutions to the equation  $z^6 = 1$  in the form  $x + yi$ . 2

- (ii) If  $\omega$  is a non-real solution to the equation  $z^6 = 1$ , 2  
 show that  $\omega^4 + \omega^2 = -1$ .

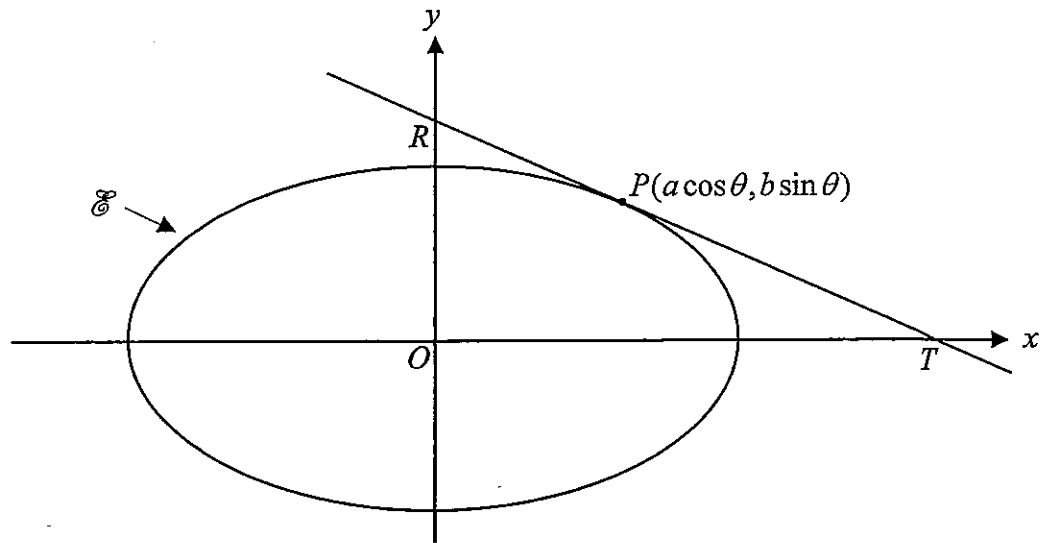
- (iii) By choosing one particular value of  $\omega$ , explain with 1  
 the aid of a diagram why  $\omega^4 + \omega^2 = -1$ .

**Question 4** (15 marks)

Marked by JSH

**Marks**

(a)



The ellipse  $E$  with equation  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  shown in the diagram above, has a tangent at the point  $P(a \cos \theta, b \sin \theta)$ . The tangent cuts the  $x$ -axis at  $T$  and the  $y$ -axis at  $R$ .

- (i) Show that the equation of the tangent at the point  $P$  is 2
- $$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1.$$

- (ii) If  $T$  is the point of intersection between the tangent at point  $P$  and one of the directrices of the ellipse, show that  $\cos \theta = e$ . 3

- (iii) Hence find the angle that the focal chord through  $P$  makes with the  $x$ -axis. 1

- (iv) Using similar triangles or otherwise, show that  $RP = e^2 RT$ . 3

- (b)  $P(ct, \frac{c}{t}), t \neq 1$  lies on the hyperbola  $xy = c^2$ . The tangent and normal at  $P$  meet the line  $y = x$  at  $T$  and  $N$  respectively. If  $O$  is the origin show that  $OT \cdot ON = 4c^2$ . Include a labelled diagram with your answer. 6

## Question 5 (15 marks)

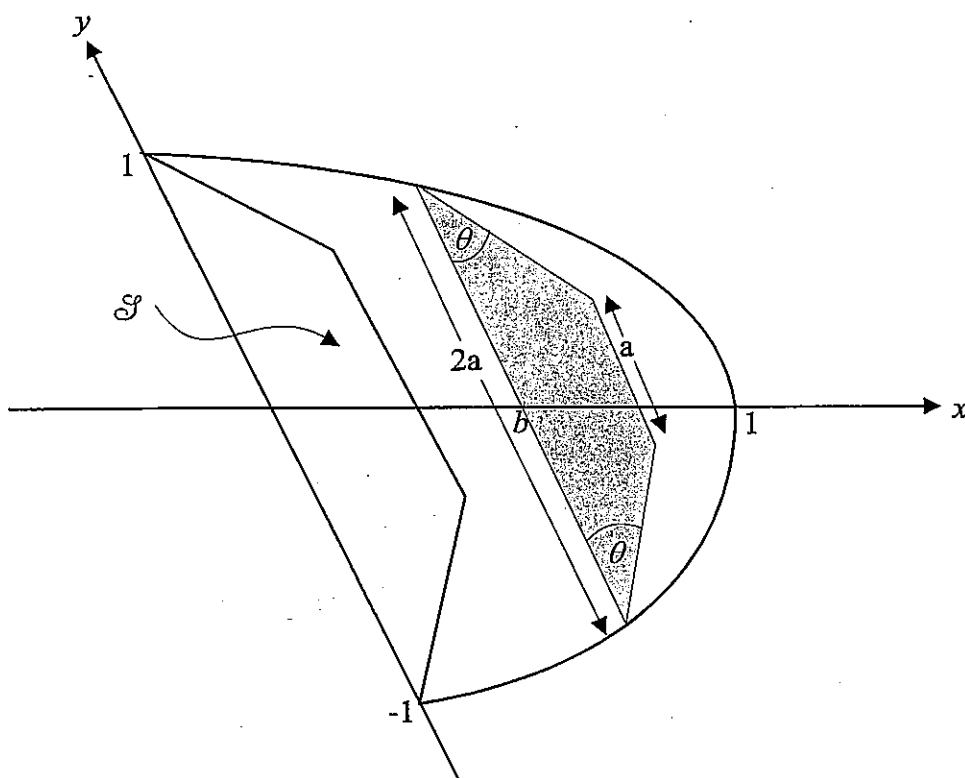
Marked by SKB

Marks

- (a) The region bounded by the curve  $x = y^2$  and the line  $x = 4$  is rotated about the line  $y = 2$ . Find the volume generated when:

- (i) Slices of thickness  $\Delta x$  are taken perpendicular to the  $x$ -axis in this region to create hollow cylindrical discs. 4
- (ii) Slices of thickness  $\Delta y$  are taken perpendicular to the  $y$ -axis in this region to create thin cylindrical shells. 4

(b)



A solid  $S$  has a semi-circular base in the  $x$ - $y$  plane with its diameter along the  $y$ -axis.

Each cross-section of the solid running perpendicular to the  $x$ - $y$  plane is a regular trapezium with its base sidelength twice that of its parallel sidelength. The angle between the base sidelength and the sides of the trapezium is  $\theta$ .

A typical cross-section taken at  $x = b$  is shown in the diagram.

- (i) Show that if  $\theta = 45^\circ$ , the area of the trapezium at  $x = b$  is  $\frac{3a^2}{4}$ . 1

- (ii) Find the volume of the solid  $S$  when  $\theta = 45^\circ$ . 2
- (iii) Find the volume of the solid,  $\mathcal{D}$ , generated when the semi-circle is rotated through an angle of  $90^\circ$  about the  $y$ -axis. 1
- (iv) Find the values of  $\theta$  for which the volume of  $S$  found in part (ii) is greater than the volume of  $\mathcal{D}$ . 3

**Question 6 (15 marks)**      Marked by JSH      **Marks**

(a) If  $1-i$  is a zero of  $P(x) = x^3 + ax^2 + bx + 6$ , where  $a, b \in \text{Real}$

(i) Evaluate  $a$  and  $b$  **4**

(ii) Hence fully factorise  $P(x)$  over the complex field. **1**

(b) (i) Use De Moivre's theorem to express  $\tan 5\theta$  in terms of powers of  $\tan \theta$ . **3**

(ii) Hence show  $x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$  has roots **2**  
 $1, \tan \frac{\pi}{20}, \tan \frac{9\pi}{20}, -\tan \frac{3\pi}{20}$  and  $-\tan \frac{7\pi}{20}$ .

(iii) By solving  $x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$  another way, show that **5**

$$\tan \frac{9\pi}{20} + \tan \frac{\pi}{20} = 2 + 2\sqrt{5} \text{ and } \tan \frac{7\pi}{20} + \tan \frac{3\pi}{20} = 2\sqrt{5} - 2.$$

## Question 7 (15 marks)

Marked by JSH

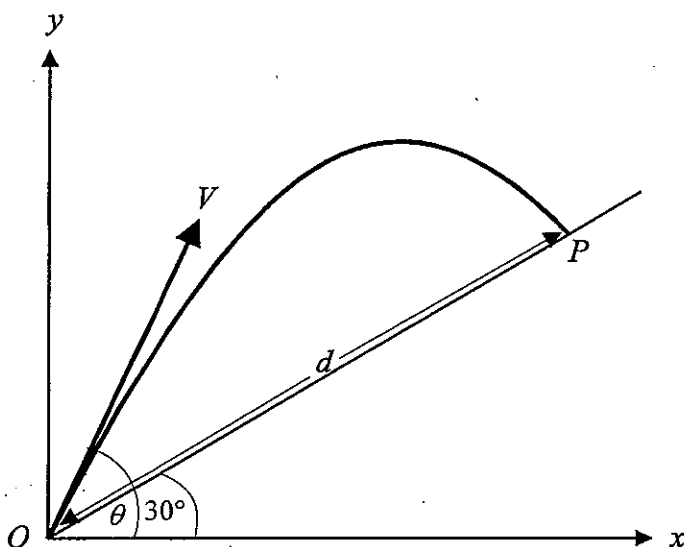
Marks

- (a) AB is a chord of a circle. X is a point on AB produced. XT is a tangent from X to the circle.

(i) Prove that  $\triangle XAT$  is similar to  $\triangle XTB$ . 2

(ii) Deduce that  $XT^2 = XA.XB$  2

(b)



The diagram above shows the path of a particle which has been projected from point  $O$  at an angle of  $\theta$  to the horizontal. The speed at which the particle was projected was  $\sqrt{g}$  m/sec where  $g$  is the acceleration due to gravity. The particle lands at point  $P$  which lies on a plane inclined at an angle of  $30^\circ$  to the horizontal. The base of this inclined plane is at  $O$  and point  $P$  lies  $d$  metres from  $O$ . The position of the particle at time  $t$  seconds is given by

$$x = \sqrt{g} t \cos \theta$$

$$\text{and } y = \sqrt{g} t \sin \theta - \frac{1}{2} g t^2$$

- (i) Show that the path of trajectory of the particle is given by 1

$$y = x \tan \theta - \frac{x^2 \sec^2 \theta}{2}.$$

- (ii) If there is only one path of trajectory for the particle to land at point  $P$ , find  $\theta$  for that path. 4

**Marks**

- (c) Find the general solutions to the equation

**6**

$$\cos 4\theta + \cos 2\theta = \sqrt{2} \cos^2 \theta + \frac{1}{\sqrt{2}} \sin 2\theta.$$

**Question 8** (15 marks)

Marked by SKB

**Marks**

- (a) (i) If  $a > 0$ ,  $b > 0$  and  $c > 0$ , show that  $a^2 + b^2 \geq 2ab$  and hence deduce that  $a^2 + b^2 + c^2 \geq ab + bc + ca$ . 2
- (ii) If  $a + b + c = 9$ , show that  $ab + bc + ca \leq 27$  and 3
- $$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{27}{abc}.$$
- (b) If  $U_1 = 1, U_2 = 5$  and  $U_n = 5U_{n-1} - 6U_{n-2}$  for  $n \geq 3$ , prove by mathematical induction that  $U_n = 3^n - 2^n$  for  $n \geq 1$ . 5
- (c) The lines  $y = 0$ ,  $3x - 4y + 3 = 0$  and  $3x + 4y - 15 = 0$  are the sides of a triangle. Find the co-ordinates of the centre of the circle inscribed in the triangle. Hence or otherwise write down the equation of the circle. 5

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$

**MATHEMATICS EXTENSION 2  
HIGHER SCHOOL CERTIFICATE  
TRIAL EXAMINATION  
SOLUTIONS  
2008**

**Question 1 (15 marks)**

(a)  $\int x \tan x^2 dx = \frac{1}{2} \int \tan u du$  where  $u = x^2$   
 (1 mark)  $\frac{du}{dx} = 2x$   
 $= \frac{1}{2} \int \frac{\sin u}{\cos u} du$   
 $= -\frac{1}{2} \ln(\cos x^2) + c$   
 (1 mark)

(b)  $\int_4^9 \frac{x}{\sqrt{x}(1+x)} dx$   $u = \sqrt{x}, \frac{du}{dx} = \frac{1}{2\sqrt{x}} \therefore 2du = \frac{dx}{\sqrt{x}}$   
 $= \int_2^3 \frac{u^2}{(1+u^2)} 2 \frac{du}{dx} dx$   $x=9, u=3$   
 $= 2 \int_2^3 \frac{u^2}{1+u^2} du$   $x=4, u=2$   
 (1 mark) for terminals  
 $= 2 \int_2^3 \left(1 - \frac{1}{1+u^2}\right) du$  (1 mark) for function  
 $= 2 \left[ u - \tan^{-1} u \right]_2^3$   
 $= 2 \left\{ (3 - \tan^{-1} 3) - (2 - \tan^{-1} 2) \right\}$   
 $= 2 - 2 \tan^{-1} 3 + 2 \tan^{-1} 2$   
 (1 mark)

$$\begin{aligned}
 \text{(c)} \quad & \int \frac{-2}{\sqrt{3+2x-x^2}} dx \\
 &= -2 \int \frac{1}{\sqrt{-(x^2-2x-3)}} dx \\
 &= -2 \int \frac{1}{\sqrt{-(x^2-2x+1-1-3)}} dx \\
 &= -2 \int \frac{1}{\sqrt{4-(x-1)^2}} dx \quad (1 \text{ mark}) \\
 &= -2 \sin^{-1} \frac{(x-1)}{2} + c \quad (1 \text{ mark})
 \end{aligned}$$

$u = x-1$   
 $\frac{du}{dx} = 1$

$$\begin{aligned}
 \text{(d)} \quad \text{(i)} \quad & \frac{2x^2+2x+5}{(x^2+2)(1-x)} = \frac{ax+b}{x^2+2} + \frac{c}{1-x} \\
 &= \frac{(ax+b)(1-x)+c(x^2+2)}{(x^2+2)(1-x)} \\
 \text{True iff} \quad & 2x^2+2x+5 = (ax+b)(1-x)+c(x^2+2) \quad (1 \text{ mark}) \\
 & \text{Put } x=1, \quad 9=3c \quad c=3 \\
 & \text{Put } x=0, \quad 5=b+2c \quad b=-1 \\
 & \text{Put } x=-1, \quad 5=(-a-1)2+3 \times 3 \quad a=1 \\
 & \text{So, } a=1, \quad b=-1, \quad c=3 \\
 & (1 \text{ mark})
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad & \text{Hence } \int \frac{2x^2+2x+5}{(x^2+2)(1-x)} dx \\
 &= \int \left( \frac{x-1}{x^2+2} + \frac{3}{1-x} \right) dx \\
 &= \int \frac{x}{x^2+2} dx - \int \frac{1}{x^2+2} dx + \int \frac{3}{1-x} dx \quad (1 \text{ mark}) \\
 &= \frac{1}{2} \ln|x^2+2| - \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} - 3 \ln|1-x| + c \\
 & \quad (or = \ln \left| \frac{\sqrt{x^2+2}}{(1-x)^3} \right| - \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + c) \quad (1 \text{ mark})
 \end{aligned}$$

$$(e) \quad \int_1^5 \frac{\ln x}{\sqrt{x}} dx = \left[ 2\sqrt{x} \ln x \right]_1^5 - \int_1^5 2\sqrt{x} \cdot \frac{1}{x} dx \quad \begin{array}{l} \text{Let } u = \ln x \quad dv = \frac{1}{\sqrt{x}} dx \\ \therefore \frac{du}{dx} = \frac{1}{x} \quad v = 2\sqrt{x} \end{array}$$

(1 mark) – first function      (1 mark) – second function

(1 mark) – correct positioning of terminals

$$\begin{aligned} &= \left( 2\sqrt{5} \ln 5 - 2 \ln 1 \right) - \int_1^5 2x^{-\frac{1}{2}} dx \\ &= 2\sqrt{5} \ln 5 - 2 \left[ 2\sqrt{x} \right]_1^5 \\ &= 2\sqrt{5} \ln 5 - 4\sqrt{5} + 4 \end{aligned}$$

(1 mark)

### Question 2 (15 marks)

$$(a) \quad I = \int_{-1}^1 \frac{\tan^{-1} x}{1+x^4} dx \quad \text{Let } f(x) = \frac{\tan^{-1} x}{1+x^4}$$

$$\therefore f(-x) = \frac{\tan^{-1}(-x)}{1+(-x)^4} = -\frac{\tan^{-1} x}{1+x^4} = -f(x)$$

$$\therefore f(x) \text{ is an odd function} \quad (1 \text{ mark})$$

$\therefore I = 0$ , as the integration of an odd function about symmetrical limits is zero. (1 mark)

$$(b) \quad I = \int_0^1 \sqrt{4-x^2} dx \quad \text{Let } x = 2 \sin \theta \therefore \frac{dx}{d\theta} = 2 \cos \theta$$

When  $x = 0, \theta = 0$  and when  $x = 1, \theta = \frac{\pi}{6}$

$$\therefore I = \int_0^{\frac{\pi}{6}} \sqrt{4-4\sin^2 \theta} \cdot 2 \cos \theta d\theta \quad (1 \text{ mark})$$

$$= 4 \int_0^{\frac{\pi}{6}} \cos^2 \theta d\theta \quad (1 \text{ mark})$$

$$= 2 \int_0^{\frac{\pi}{6}} 1 + \cos 2\theta d\theta, \text{ using } \cos 2\theta = 2\cos^2 \theta - 1 \text{ and } \cos^2 \theta = \frac{1}{2}[1 + \cos 2\theta]$$

$$= 2 \left[ \theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{6}} \quad (1 \text{ mark})$$

$$= 2 \left[ \frac{\pi}{6} + \frac{\sqrt{3}}{4} \right] \text{ or } \frac{2\pi + 3\sqrt{3}}{6} \text{ or } \frac{\pi}{3} + \frac{\sqrt{3}}{2} \quad (1 \text{ mark})$$

$$\begin{aligned}
 \text{(c)} \quad I &= \int_0^{2\pi} \frac{x \cos x}{1 + \sin^2 x} dx \\
 &= \int_0^{2\pi} \frac{(2\pi - x) \cos(2\pi - x)}{1 + \sin^2(2\pi - x)} dx, \text{ using the given result (1 mark)}
 \end{aligned}$$

$$\begin{aligned}
 &= \int_0^{2\pi} \frac{2\pi \cos x}{1 + \sin^2 x} dx - \int_0^{2\pi} \frac{x \cos x}{1 + \sin^2 x} dx \\
 \therefore 2I &= \int_0^{2\pi} \frac{2\pi \cos x}{1 + \sin^2 x} dx
 \end{aligned}$$

$$\therefore I = \pi \int_0^{2\pi} \frac{\cos x}{1 + \sin^2 x} dx \quad \text{(1 mark)} \quad \text{Let } u = \sin x, \therefore \frac{du}{dx} = \cos x$$

When  $x = 0, u = 0$  and when  $x = 2\pi, u = 0$

$$\therefore I = \pi \int_0^0 \frac{du}{1 + u^2} = 0, \text{ as the integration about the same limits is zero. (1 mark)}$$

$$\text{(d)} \quad \text{(i)} \quad I_n = \int_0^1 x(x^2 - 1)^n dx \quad n = 0, 1, 2, \dots$$

$$= \left[ \frac{x^2}{2} (x^2 - 1)^n \right]_0^1 - \int_0^1 \frac{x^2}{2} \times n(x^2 - 1)^{n-1} \times 2x dx \quad \text{(1 mark)}$$

$$= \frac{1}{2} \times 0 - 0 - n \int_0^1 x^3 (x^2 - 1)^{n-1} dx$$

$$= -n \int_0^1 \frac{x^3 (x^2 - 1)^n}{x^2 - 1} dx$$

$$= -n \int_0^1 \frac{[x(x^2 - 1) + x](x^2 - 1)^n}{x^2 - 1} dx$$

$$= -n \int_0^1 \left\{ x(x^2 - 1)^n + \frac{x}{x^2 - 1} (x^2 - 1)^n \right\} dx \quad \text{(1 mark)}$$

$$I_n = -n \int_0^1 x(x^2 - 1)^n dx - n \int_0^1 x(x^2 - 1)^{n-1} dx$$

$$I_n = -nI_n - nI_{n-1}$$

$$(1 + n)I_n = -nI_{n-1}$$

$$\text{So } I_n = \frac{-n}{n+1} I_{n-1} \text{ for } n \geq 1 \text{ as required. (1 mark)}$$

(ii) Method 1 – “Hence”

$$I_n = \frac{-n}{n+1} I_{n-1} \quad \text{for } n \geq 1$$

$$= \frac{-n}{n+1} \frac{-n+1}{n} \frac{-n+2}{n-1} \dots \frac{-3}{4} \frac{-2}{3} \frac{-1}{2} I_0$$

$$\text{Now } I_0 = \int_0^1 x(x^2-1)^0 dx$$

$$= \left[ \frac{x^2}{2} \right]_0^1$$

$$= \frac{1}{2}$$

(1 mark)

$$\text{So } I_n = \frac{-n}{n+1} \frac{-n+1}{n} \frac{-n+2}{n-1} \dots \frac{-3}{4} \frac{-2}{3} \frac{-1}{2} \frac{1}{2} \text{ for } n \geq 0$$

$$= (-1)^n \frac{n}{n+1} \frac{n-1}{n} \frac{n-2}{n-1} \dots \frac{3}{4} \frac{2}{3} \frac{1}{2} \frac{1}{2}$$

$$= (-1)^n \frac{1}{n+1} \cdot \frac{1}{2} \text{ (The other terms cancel.)}$$

(1 mark)

$$\text{So } I_n = \frac{(-1)^n}{2(n+1)}, \quad n \geq 0 \quad \text{as required.}$$

**OR**

$$I_n = \frac{-n}{n+1} I_{n-1} \quad \text{for } n \geq 1$$

$$I_n = \frac{-n}{n+1} \int_0^1 x(x^2-1)^{n-1} dx$$

$$= \frac{-n}{n+1} \frac{1}{2} \int_0^1 2x(x^2-1)^{n-1} dx$$

$$= \frac{-n}{n+1} \frac{1}{2} \left[ \frac{(x^2-1)^n}{n} \right]_0^1$$

(2 marks)

$$= \frac{-n}{n+1} \frac{1}{2} \left[ 0 - \frac{(-1)^n}{n} \right]$$

$$\text{So } I_n = \frac{(-1)^n}{2(n+1)}, \quad n \geq 0 \quad \text{as required.}$$

(ii) Method 2 – “Otherwise”

$$\begin{aligned}
 I_n &= \int_0^1 x(x^2 - 1)^n dx \text{ for } n = 0, 1, 2, \dots \\
 &= \frac{1}{2} \int_0^1 2x(x^2 - 1)^n dx \\
 &= \frac{1}{2} \left[ \frac{(x^2 - 1)^{n+1}}{n+1} \right]_0^1 \quad (1 \text{ mark}) \\
 &= \frac{1}{2(n+1)} (0 - (-1)^{n+1}) \\
 &= \frac{-1(-1)^{n+1}}{2(n+1)} \\
 &= \frac{(-1)^n}{2(n+1)} \text{ for } n \geq 0 \text{ as required} \quad (1 \text{ mark})
 \end{aligned}$$

$$(iii) \quad I_0 = \frac{1}{2}, I_1 = \frac{-1}{4}, I_2 = \frac{1}{6}, I_3 = \frac{-1}{8}, I_4 = \frac{1}{10}, I_5 = \frac{-1}{12}$$

Clearly  $I_{2n} > 0$  and  $I_{2n+1} < 0$ So  $I_{2n} > I_{2n+1}$ 

$$\begin{aligned}
 \text{Alternatively, from (ii), } I_{2n} &= \frac{(-1)^{2n}}{2(2n+1)} \\
 &= \frac{1}{2(2n+1)} \\
 &> 0 \\
 I_{2n+1} &= \frac{(-1)^{2n+1}}{2((2n+1)+1)} \\
 &= \frac{(-1)}{4(n+1)} \\
 &< 0
 \end{aligned}$$

So  $I_{2n} > I_{2n+1}$ .

(1 mark)

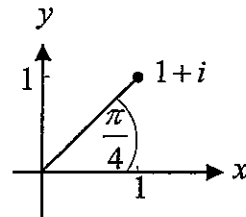
**Question 3 (15 marks)**

$$\begin{aligned}
 (a) \quad (i) \quad z\bar{w} &= (3-i)(2-4i) \\
 &= 6 - 12i - 2i - 4 \\
 &= 2 - 14i \quad (1 \text{ mark})
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \frac{z}{w} &= \frac{3-i}{2+4i} \times \frac{2-4i}{2-4i} \\
 &= \frac{2-14i}{20} \\
 &= \frac{1}{10} - \frac{7}{10}i
 \end{aligned}$$

**(1 mark)**

(b) (i) From the diagram,  
 $1+i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$



**(1 mark)** for correct modulus  
**(1 mark)** for correct argument

(ii)  $(1+i)^n + (1-i)^n = 0$

Hence

$$\left(\sqrt{2}cis\frac{\pi}{4}\right)^n + \left(\sqrt{2}cis\frac{-\pi}{4}\right)^n = 0 \quad (1 \text{ mark})$$

$$(\sqrt{2})^n \left(\cos\frac{\pi n}{4} + i\sin\frac{\pi n}{4}\right) + (\sqrt{2})^n \left(\cos\frac{\pi n}{4} - i\sin\frac{\pi n}{4}\right) = 0$$

$$\cos\frac{\pi n}{4} = 0 \quad (1 \text{ mark})$$

$$\frac{\pi n}{4} = \frac{(2k+1)\pi}{2}$$

where  $k = 0, 1, 2, \dots$

$$n = 2(2k+1)$$

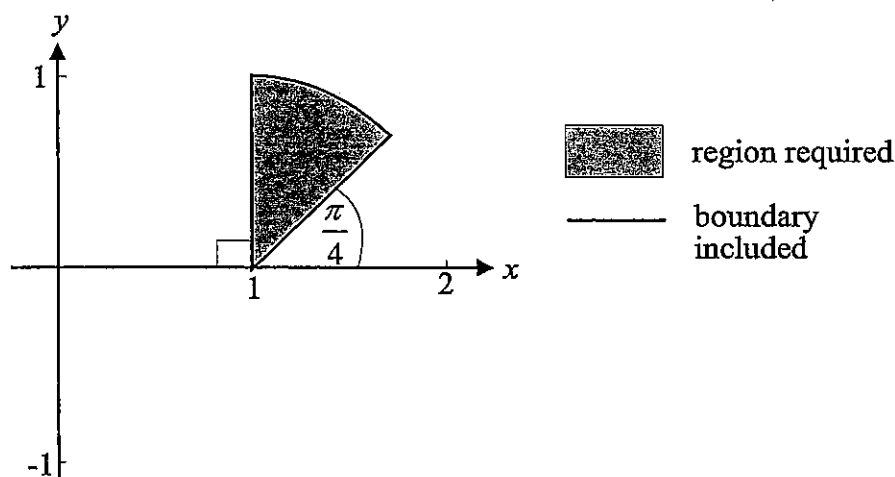
(1 mark)

- (c) The inequality  $|z-1| \leq 1$  corresponds to a disc with centre at  $(1,0)$  and radius 1.

The inequality  $\frac{\pi}{4} \leq \arg(z-1) \leq \frac{\pi}{2}$  corresponds to a wedge with vertex  $(1,0)$ .

(1 mark)

The region where both these inequalities hold is shown in the diagram below.



(1 mark)

- (d) Point  $P$  corresponds to the complex number  $z$ . Point  $Q$  is obtained by rotating point  $P$  **clockwise** through an angle of  $\frac{\pi}{2}$  and reducing it by a factor of  $\frac{1}{3}$ .

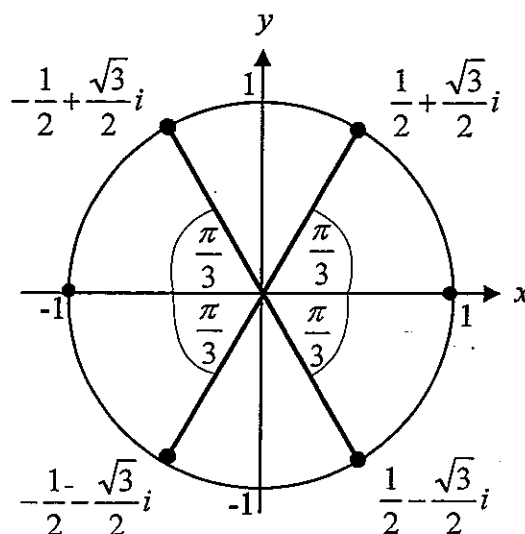
So, point  $Q$  corresponds to the complex number  $\frac{-i}{3}z$ .

(1 mark)

(e) (i)  $z^6 = 1$

We are looking for the sixth roots of unity. We know that one root is 1 and another is  $-1$ . The 6 roots of unity are evenly spaced around the circumference of a circle of radius 1 unit.

So, the other four must be  $\text{cis}\frac{\pi}{3}$ ,  $\text{cis}\frac{2\pi}{3}$ ,  $\text{cis}\frac{4\pi}{3}$  and  $\text{cis}\frac{5\pi}{3}$ .



(1 mark)

So, the six roots are

$$\pm 1, +\frac{1}{2} \pm \frac{\sqrt{3}}{2}i, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i. \quad (1 \text{ mark})$$

(ii)  $z^6 - 1 = 0$

$$(z^3 - 1)(z^3 + 1) = 0$$

$$(z - 1)(z^2 + z + 1)(z + 1)(z^2 - z + 1) = 0 \quad (1 \text{ mark})$$

The two real roots of the equation are revealed by the factors  $(z - 1)$  and  $(z + 1)$ . The four non-real roots are revealed by the factors  $(z^2 + z + 1)$  and  $(z^2 - z + 1)$ .

$$\text{So } (\omega^2 + \omega + 1)(\omega^2 - \omega + 1) = 0$$

$$\omega^4 + \omega^2 + 1 = 0$$

$$\text{So } \omega^4 + \omega^2 = -1 \quad \text{as required.} \quad (1 \text{ mark})$$

(iii) Let  $\omega = \text{cis}\left(\frac{\pi}{3}\right)$

Now,  $\omega^4 = \text{cis}\frac{4\pi}{3}$  (De Moivre)

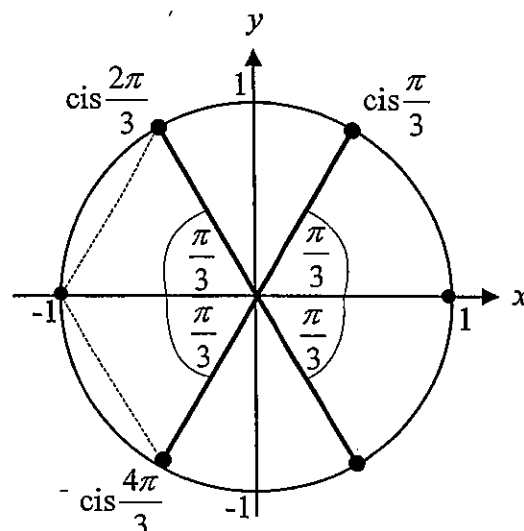
$\omega^2 = \text{cis}\frac{2\pi}{3}$  (De Moivre)

$\text{cis}\frac{4\pi}{3} + \text{cis}\frac{2\pi}{3} = -1$  by adding

the two complex numbers  $\text{cis}\frac{4\pi}{3}$  and  $\text{cis}\frac{2\pi}{3}$ .

(Note, any of the four possible values of  $\omega$  could have been chosen here to illustrate that  $\omega^4 + \omega^2 = -1$ .)

(1 mark)



**Question 4 (15 marks)**

(a) (i)  $P$  is the point  $(a \cos \theta, b \sin \theta)$

$x = a \cos \theta$        $y = b \sin \theta$

$\frac{dx}{d\theta} = -a \sin \theta$        $\frac{dy}{d\theta} = b \cos \theta$

$\frac{dy}{dx} = -\frac{b \cos \theta}{a \sin \theta}$

(1 mark)

Equation of tangent is

$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$

$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$

$bx \cos \theta + ay \sin \theta = ab(\sin^2 \theta + \cos^2 \theta)$

$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$

(1 mark)

as required.

- (ii) If  $T$  is the point of intersection between the tangent found in part (i) and one of the directrices of the ellipse, then  $T$  has the coordinates  $\left(\frac{a}{e}, 0\right)$ . **(1 mark)**

From (i), the gradient of the tangent at  $P$  is  $\frac{-b \cos \theta}{a \sin \theta}$ . So using the coordinates of points  $P$  and  $T$  we have

$$\frac{b \sin \theta - 0}{a \cos \theta - \frac{a}{e}} = \frac{-b \cos \theta}{a \sin \theta} \quad \text{(1 mark)}$$

$$\frac{eb \sin \theta}{ae \cos \theta - a} = \frac{-b \cos \theta}{a \sin \theta}$$

$$be \sin \theta \times a \sin \theta = -b \cos \theta (ae \cos \theta - a)$$

$$abe \sin^2 \theta = -abe \cos^2 \theta + ab \cos \theta$$

$$abe(\sin^2 \theta + \cos^2 \theta) = ab \cos \theta$$

$$\cos \theta = e \quad \text{as required.}$$

**(1 mark)**

- (iii) Since  $\cos \theta = e$ , the  $x$ -coordinate of  $P$  which is  $a \cos \theta = ae$ . So the focal chord through  $P$  makes an angle of  $90^\circ$  with the  $x$ -axis. **(1 mark)**

- (iv) The equation of the tangent through  $P$  is

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

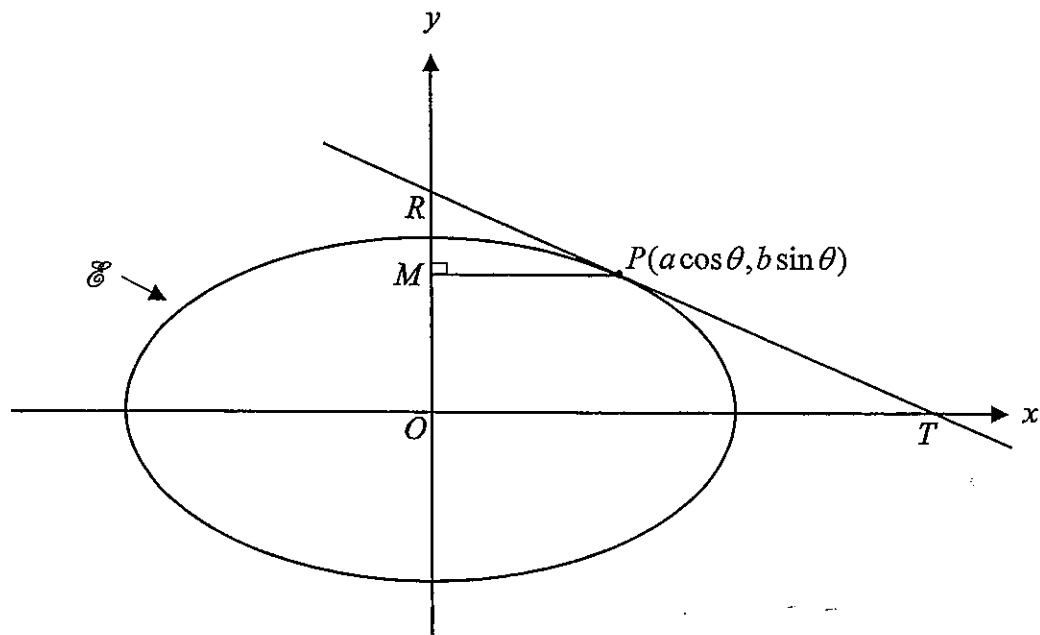
when  $x = 0$ ,

$$y \sin \theta = b$$

$$y = \frac{b}{\sin \theta}$$

$R$  is the point  $\left(0, \frac{b}{\sin \theta}\right)$

**(1 mark)**



Let  $M$  be the point on the  $y$ -axis such that  $PM$  is perpendicular to the  $y$ -axis.

$M$  is the point  $(0, b \sin \theta)$

Now since  $\triangle ROT$  is similar to  $\triangle RMP$ ,

$$\frac{RP}{RT} = \frac{RM}{RO} \quad (1 \text{ mark})$$

$$= \frac{\frac{b}{\sin \theta} - b \sin \theta}{\frac{b}{\sin \theta}}$$

$$= \frac{b - b \sin^2 \theta}{\sin \theta} \times \frac{\sin \theta}{b}$$

$$= 1 - \sin^2 \theta$$

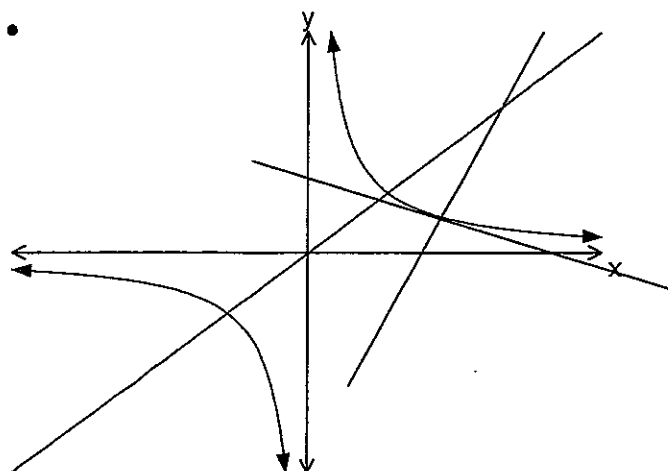
$$= \cos^2 \theta$$

$$= e^2 \quad (\text{from part (ii)})$$

So  $RP = e^2 RT$  as required.

(1 mark)

(b) (1 mark)



: x Intercept (0, 0)

$$xy = c^2 \therefore y = c^2 x^{-1} \text{ and } \frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$\text{At } P(ct, \frac{c}{t}) \quad \frac{dy}{dx} = \frac{-c^2}{c^2 t^2} = \frac{-1}{t^2} = m_{\text{tangent}}; m_{\text{normal}} = t^2 \quad (1 \text{ mark})$$

Now the equation of the tangent at P is:

$$y - \frac{c}{t} = \frac{-1}{t^2}(x - ct)$$

$$\therefore t^2 y - ct = -x + ct$$

$$\therefore x + t^2 y = 2ct$$

$$\text{At T } y = x \therefore x + t^2 x = 2ct \therefore x = \frac{2ct}{1+t^2}$$

$$\Rightarrow T = (\frac{2ct}{1+t^2}, \frac{2ct}{1+t^2}) \quad (1 \text{ mark})$$

Now the equation of the normal at P is:

$$y - \frac{c}{t} = t^2(x - ct)$$

$$\therefore t^3 x - ty = c(t^4 - 1)$$

$$\text{At N } y = x \therefore t^3 x - tx = c(t^4 - 1) \therefore x = \frac{c(t^4 - 1)}{t(t^2 - 1)} = \frac{c(t^2 + 1)}{t}$$

$$\Rightarrow N = (\frac{c(t^2 + 1)}{t}, \frac{c(t^2 + 1)}{t}) \quad (1 \text{ mark})$$

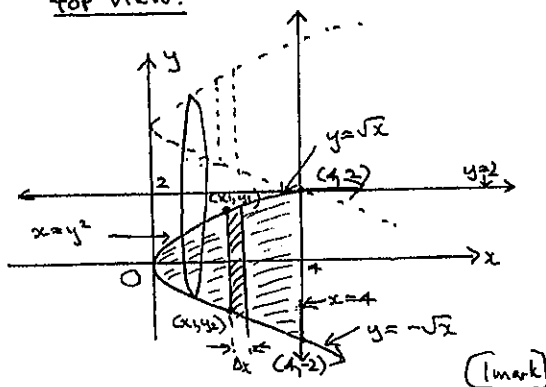
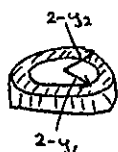
$$\text{Now } OT \cdot ON = \sqrt{(\frac{2ct}{1+t^2})^2 + (\frac{2ct}{1+t^2})^2} \cdot \sqrt{(\frac{c(t^2 + 1)}{t})^2 + (\frac{c(t^2 + 1)}{t})^2}$$

$$= \sqrt{2}(\frac{2ct}{1+t^2}) \cdot \sqrt{2}(\frac{c(t^2 + 1)}{t})$$

$$= 4c^2$$

(2 marks)

5 (a)

TOP VIEW:Take slice of thickness  $\Delta x \perp$  to  $x$ -axisSIDE VIEW:

$$y_1 = \sqrt{x}, R_2 = 2 - y_2$$

$$y_2 = -\sqrt{x}, R_1 = 2 - y_1$$

Area of cross-sectional slice,  $A(x) = \pi(R_2^2 - R_1^2)$ 

$$\therefore A(x) = \pi((2 - \sqrt{x})^2 - (2 + \sqrt{x})^2)$$

$$= \pi(4 - 4\sqrt{x} + x - (4 + 4\sqrt{x} + x))$$

$$= 8\pi\sqrt{x} \quad (1 \text{ mark})$$

$$\text{Volume, } \Delta V, \text{ of each slice} = A(x)\Delta x$$

$$\text{Now total volume} = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^4 A(x)\Delta x$$

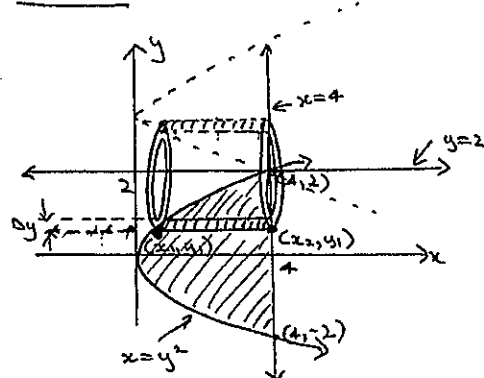
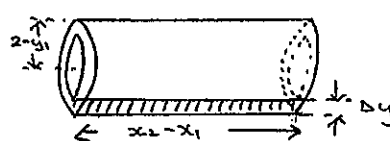
$$= 8\pi \int_0^4 x^{\frac{1}{2}} dx \quad (1 \text{ mark})$$

$$= 8\pi \left[ \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^4$$

$$= \frac{16\pi}{3} [4^{\frac{3}{2}} - 0]$$

$$= \frac{128\pi}{3} \text{ units}^3 \quad (1 \text{ mark})$$

(b)

TOP VIEW:Take slice of thickness  $\Delta y \perp$  to  $y$ -axisSIDE VIEW:Each slice when rotated about the line  $y = 2$  generates a thin cylindrical shell of area  $A(y) = 2\pi rh$ .Volume,  $\Delta V$ , of each slice  $= A(y)\Delta y$ 

$$\text{Now total volume} = \lim_{\Delta y \rightarrow 0} \sum_{y=-2}^2 A(y)\Delta y$$

$$\text{where } A(y) = 2\pi(2-y)(4-y^2) \quad (1 \text{ mark})$$

$$\therefore \text{Total volume} = 2\pi \int_{-2}^2 (8 - 4y - 2y^3 + y^5) dy$$

$$(1 \text{ mark})$$

$$= 2\pi \left[ 8y - 2y^2 - \frac{2y^4}{4} + \frac{y^6}{6} \right]_{-2}^2$$

$$= 2\pi \left[ (16 - 8 - \frac{16}{3} + \frac{64}{6}) - (-16 - 8 + \frac{16}{3} + \frac{64}{6}) \right]$$

$$= 2\pi \left[ 32 - \frac{16}{3} \times 2 \right]$$

$$(1 \text{ mark}) \quad = \frac{128\pi}{3} \text{ units}^3$$

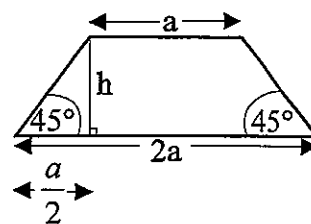
(c) (i)  $\tan 45^\circ = h \div \frac{a}{2}$

$$h = \frac{a}{2}$$

Area of trapezium at  $x = b$  is given by

$$\begin{aligned} & \frac{1}{2}(a+2a)\frac{a}{2} \\ &= \frac{3a^2}{4} \end{aligned}$$

(1 mark)



(ii) Consider a slice of width  $\Delta x$

$$\Delta V = \frac{3a^2}{4} \Delta x \quad \text{where } 0 \leq x \leq 1$$

$$= \frac{3}{4} y^2 \Delta x$$

$$= \frac{3}{4} (1-x^2) \Delta x \quad \text{since } x^2 + y^2 = 1 \quad (1 \text{ mark})$$

$$\text{So Volume} = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 \frac{3}{4} (1-x^2) \Delta x$$

$$= \frac{3}{4} \int_0^1 (1-x^2) dx$$

$$= \frac{3}{4} \left[ x - \frac{x^3}{3} \right]_0^1$$

$$= \frac{3}{4} \left( \frac{2}{3} \right)$$

$$= \frac{1}{2} \text{ unit}^3$$

(1 mark)

(iii) The solid generated when the semicircular base of  $\mathcal{D}$  is rotated through an angle of  $90^\circ$  is a quarter of a sphere. Hence the volume is

$$\frac{4}{3} \pi r^3 \div 4. \text{ So required volume is } \frac{\pi}{3} \text{ since } r = 1.$$

(1 mark)

(iv) From part (i), area of a trapezium at  $x = b$  is given by  $\frac{1}{2}(a+2a)h$  where  $h$  is the height of the trapezium.

$$\begin{aligned}\text{Now } \tan \theta &= h \div \frac{a}{2} \\ &= \frac{2h}{a}\end{aligned}$$

So area of trapezium at  $x = b$

$$\begin{aligned}&= \frac{3a}{2} \times \frac{a}{2} \tan \theta \\ &= \frac{3a^2}{4} \tan \theta\end{aligned}$$

$$\begin{aligned}\text{So Volume} &= \frac{3}{4} \tan \theta \int_0^1 (1-x^2) dx \quad \text{where } \theta \text{ is constant for a particular solid} \\ &= \frac{1}{2} \tan \theta \quad \text{(1 mark)}\end{aligned}$$

We want to find  $\theta$  such that

$$\begin{aligned}\frac{1}{2} \tan \theta &> \frac{\pi}{3} \\ \tan \theta &> \frac{2\pi}{3} \\ \theta &> 64^\circ 29' \quad \text{(1 mark)}\end{aligned}$$

However  $\theta < 90^\circ$  since we have a trapezium.

So we require  $64^\circ 29' < \theta < 90^\circ$ . (1 mark)

### Question 6 (15 marks)

- (a) (i) As the coefficients of  $P(x)$  are real then  $1+i$  is a further root of  $P(x)$ .

$$\begin{aligned}\therefore P(x) &= (x-1+i)(x-1-i)R(x) \\ &= ([x-1]^2 - i^2)R(x) \quad \text{(1mark)} \\ &= (x^2 - 2x + 2)R(x)\end{aligned}$$

As  $1-i$  is a root then  $P(1-i) = 0$

$$\begin{aligned}\therefore (1-i)^3 + a(1-i)^2 + b(1-i) + 6 &= 0 \\ \therefore 1 + 3(-i) + 3(-i)^2 + (-i)^3 + a(1-2i+i^2) + b-bi + 6 &= 0 \quad \text{(1mark)} \\ \therefore 1-3i-3+i+a-2ai-a+b-bi+6 &= 0 \\ \therefore 4+b+i(-2-2a-b) &= 0\end{aligned}$$

Equating real and imaginary parts:

$$4+b=0 \dots\dots\dots(1) \quad \text{(1mark)}$$

$$-2-2a-b=0 \dots\dots\dots(2)$$

$$\text{From (1) } b = -4 \text{ sub. into (2) } \therefore a = 1 \quad \text{(1mark)}$$

$$\begin{aligned}\text{(ii) } \therefore P(x) &= x^3 + x^2 - 4x + 6 \\ &= (x^2 - 2x + 2)(x+3) \\ &= (x-1+i)(x-1-i)(x+3) \text{ over the complex field.} \\ &\quad \text{(1mark)}\end{aligned}$$

(b)  $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$  (De Moivre)

$$\begin{aligned}
 (\cos \theta + i \sin \theta)^5 &= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3 \\
 &\quad + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5 \\
 &= \cos^5 \theta + i 5 \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - i 10 \cos^2 \theta \sin^3 \theta \\
 &\quad + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta
 \end{aligned}$$

(1 mark)

By equating real and imaginary parts,

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

(1 mark)

$$\begin{aligned}
 \tan 5\theta &= \frac{\sin 5\theta}{\cos 5\theta} \\
 &= \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta} \\
 &= \frac{\frac{5 \sin \theta}{\cos \theta} - \frac{10 \sin^3 \theta}{\cos^3 \theta} + \frac{\sin^5 \theta}{\cos^5 \theta}}{1 - \frac{10 \sin^2 \theta}{\cos^2 \theta} + \frac{5 \sin^4 \theta}{\cos^4 \theta}}
 \end{aligned}$$

$$\text{So } \tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

(1 mark)

(ii)

Let  $x = \tan \theta$  and  $\tan 5\theta = 1$

$$\text{So, } \tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

becomes  $\frac{5x - 10x^3 + x^5}{1 - 10x^2 + 5x^4} = 1$

$$\text{So, } x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$$

For  $\tan 5\theta = 1$

$$5\theta = n\pi + \frac{\pi}{4}, \quad n \text{ is an integer}$$

$$\theta = \frac{n\pi}{5} + \frac{\pi}{20}$$

$$n = 0, \theta = \frac{\pi}{20}$$

$$n = 1, \theta = \frac{\pi}{4}$$

$$n = -1, \theta = \frac{-3\pi}{20}$$

$$n = 2, \theta = \frac{9\pi}{20}$$

$$n = -2, \theta = \frac{-7\pi}{20}$$

$$\text{So, } x = \tan \frac{\pi}{4} = 1, \tan \frac{\pi}{20}, \tan \frac{9\pi}{20}, \tan \left( \frac{-3\pi}{20} \right) = -\tan \frac{3\pi}{20} \text{ and } \tan \left( \frac{-7\pi}{20} \right) = -\tan \frac{7\pi}{20}$$

(2 marks)

(iii)

$$\text{let } P(x) = x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$$

$$(x-1)(x^4 - 4x^3 - 14x^2 - 4x + 1) = 0 \quad (1 \text{ mark})$$

$$x^4 - 4x^3 - 14x^2 - 4x + 1 = x^2 \left( x^2 - 4x - 14 - \frac{4}{x} + \frac{1}{x^2} \right) = 0$$

$$\text{So, } x^2 \left( x^2 + \frac{1}{x^2} - 4 \left( x + \frac{1}{x} \right) - 14 \right) = 0$$

$$\text{but } x^2 \neq 0 \text{ so, } \left( x + \frac{1}{x} \right)^2 - 4 \left( x + \frac{1}{x} \right) - 16 = 0$$

$$\begin{aligned} \text{since } P(0) \neq 0, \quad x + \frac{1}{x} &= \frac{4 \pm \sqrt{16 - 4(1)(-16)}}{2} \\ &= \frac{4 \pm \sqrt{80}}{2} \\ &= \frac{4 \pm 4\sqrt{5}}{2} \end{aligned}$$

$$\text{So, } x + \frac{1}{x} = 2 + 2\sqrt{5}, \text{ or } x + \frac{1}{x} = 2 - 2\sqrt{5}$$

(2 marks)

Now 
$$\tan \theta = \cot \left( \frac{\pi}{2} - \theta \right) = \frac{1}{\tan \left( \frac{\pi}{2} - \theta \right)},$$

so, if  $x = \tan \frac{\pi}{20}$  then  $\tan \frac{\pi}{20} + \frac{1}{\tan \frac{\pi}{20}} = 2 + 2\sqrt{5}$

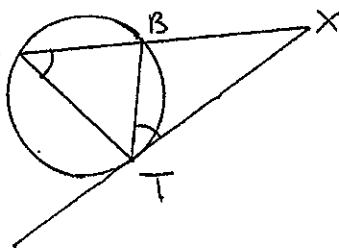
and hence  $\tan \frac{\pi}{20} + \tan \frac{9\pi}{20} = 2 + 2\sqrt{5}$

Similarly, if  $x = -\tan \frac{3\pi}{20}$  then  $-\tan \frac{3\pi}{20} - \frac{1}{\tan \frac{3\pi}{20}} = 2 - 2\sqrt{5}$

and hence  $\tan \frac{3\pi}{20} + \tan \frac{7\pi}{20} = 2\sqrt{5} - 2$  **(2 marks)**

Question 7

(a) (i) A

In  $\Delta$ s XAT and XTB:

$$\angle XAT = \angle XTB$$

[Angle between tangent and chord at point of contact equals angle in alternate segment] [1 mark]

$\angle X$  is common

$$\angle XTA = \angle XBT \quad \begin{array}{l} \text{[Remaining } \angle\text{s} \\ \text{are equal;} \\ \angle \text{sum of } \Delta = 180^\circ \end{array}$$

$\therefore$  As  $\Delta$ s are equiangular

$$\therefore \Delta XAT \sim \Delta XTB \quad [1 \text{ mark}]$$

(ii) Now  $\frac{XA}{XT} = \frac{XT}{XB}$  [Corresponding sides of similar  $\Delta$ s are in the same ratio.] [1 mark]

$$\therefore XT^2 = XA \cdot XB. \quad [1 \text{ mark}]$$

(b)

$$(i) \quad x = \sqrt{g} t \cos \theta$$

$$\text{So} \quad t = \frac{x}{\sqrt{g} \cos \theta}$$

$$\text{In} \quad y = \sqrt{g} t \sin \theta - \frac{1}{2} g t^2$$

$$\text{becomes } y = \frac{\sqrt{g} x \sin \theta}{\sqrt{g} \cos \theta} - \frac{1}{2} g \frac{x^2}{g \cos^2 \theta}$$

$$\text{So,} \quad y = x \tan \theta - \frac{x^2 \sec^2 \theta}{2}$$

(1 mark)

(ii) From (i) we have

$$y = x \tan \theta - \frac{x^2 \sec^2 \theta}{2}$$

$$= x \tan \theta - \frac{x^2}{2} (1 + \tan^2 \theta)$$

At point P,  $x = d \cos 30^\circ$  and  $y = d \sin 30^\circ$ 

$$= \frac{\sqrt{3}d}{2} \quad = \frac{d}{2}$$

(1 mark)

$$\text{So} \quad \frac{d}{2} = \frac{\sqrt{3}d}{2} \tan \theta - \frac{3d^2}{8} (1 + \tan^2 \theta)$$

$$4d = 4\sqrt{3}d \tan \theta - 3d^2 - 3d^2 \tan^2 \theta$$

$$3d^2 \tan^2 \theta - 4\sqrt{3}d \tan \theta + 3d^2 + 4d = 0$$

(1 mark)

We have a quadratic in  $\tan \theta$ .

$$\text{So} \quad \Delta = 48d^2 - 4 \times 3d^2 (3d^2 + 4d)$$

$$= 48d^2 - 12d^2 (3d^2 + 4d)$$

$$= 12d^2 (4 - 3d^2 - 4d)$$

If there is one path of trajectory for the particle to land at point P then  $\Delta = 0$ .

So  $12d^2(4-3d^2-4d)=0$   
 $4-3d^2-4d=0$  ( $12d^2 \neq 0$ )  
 $(-3d+2)(d+2)=0$   
 $d=\frac{2}{3}$  or  $d=-2$  reject this since  $d > 0$   
 So  $d=\frac{2}{3}$  (1 mark)  
 So we have,  $\frac{4}{3}\tan^2\theta - \frac{8\sqrt{3}}{3}\tan\theta + 4 = 0$   
 So  $\tan\theta = \left(\frac{8\sqrt{3}}{3} \pm \sqrt{0}\right) \div \frac{8}{3}$   
 $= \sqrt{3}$   
 $\theta = 60^\circ$  (1 mark)

(c)

$$\cos 4\theta + \cos 2\theta = \sqrt{2}\cos^2\theta + \frac{1}{\sqrt{2}}\sin 2\theta$$

$$2\cos 3\theta \cos \theta = \sqrt{2}\cos^2\theta + \sqrt{2}\sin\theta \cos\theta$$
(1 mark)

$$\sqrt{2}\cos 3\theta \cos \theta = \cos\theta(\cos\theta + \sin\theta)$$

$$= \cos\theta\left(\sqrt{2}\cos\left(\theta - \frac{\pi}{4}\right)\right)$$
(1 mark)

$$\text{So } \cos 3\theta \cos \theta = \cos\theta \cos\left(\theta - \frac{\pi}{4}\right)$$

$$\cos\theta = 0 \text{ or } \cos 3\theta = \cos\left(\theta - \frac{\pi}{4}\right)$$
(1 mark)

$$\theta = 2n\pi \pm \frac{\pi}{2} \text{ or } 3\theta = 2n\pi \pm \left(\theta - \frac{\pi}{4}\right) \quad n \text{ is an integer}$$

$$2\theta = 2n\pi - \frac{\pi}{4} \quad \text{or} \quad 4\theta = 2n\pi + \frac{\pi}{4}$$

So,  $\theta = 2n\pi \pm \frac{\pi}{2}$  (1 mark) or  $\theta = n\pi - \frac{\pi}{8}$  (1 mark) or  $\theta = \frac{n\pi}{2} + \frac{\pi}{16}$  (1 mark)

**Question 8 (15 marks)**

(a) (i)  $(a-b)^2 \geq 0$

$$a^2 + b^2 - 2ab \geq 0$$

$$a^2 + b^2 \geq 2ab \quad (\text{equality iff } a = b) \quad (1 \text{ mark})$$

Similarly,  $b^2 + c^2 \geq 2bc$

$$a^2 + c^2 \geq 2ac$$

By addition,  $2(a^2 + b^2 + c^2) \geq 2(ab + bc + ca)$

$$a^2 + b^2 + c^2 \geq ab + bc + ca \quad (\text{equality iff } a = b = c)$$

(1 mark)

(ii)  $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + ac + bc)$

Since  $a^2 + b^2 + c^2 \geq ab + bc + ca$

$$(a+b+c)^2 \geq ab + bc + ca + 2(ab + ac + bc)$$

$$(a+b+c)^2 \geq 3(ab + bc + ca)$$

Since  $a + b + c = 9$ ,

$$81 \geq 3(ab + bc + ca)$$

$$ab + bc + ca \leq 27 \quad (\text{equality iff } a = b = c) \quad (2 \text{ marks})$$

$$\frac{1}{abc}(ab + bc + ca) \leq \frac{27}{abc}$$

$$\frac{1}{c} + \frac{1}{a} + \frac{1}{b} \leq \frac{27}{abc}$$

$$\text{ie } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{27}{abc} \quad \text{as required} \quad (1 \text{ mark})$$

8(b)

Step 1: When  $n=1$   $U_1 = 3^1 - 2^1 = 1$   
which is true  
 $n=2$   $U_2 = 3^2 - 2^2 = 5$   
which is true  
 $n=3$   $U_3 = 5U_2 - 6U_1$   
 $= 25 - 6$   
 $= 19$   
 $= 3^3 - 2^3$   
 which is true

[2 marks]

 $\therefore$  it is true for  $n=1, 2$  and  $3$ .

Step 2: Assume it is true for  $n=k$   
 $(k \leq n, k \in \mathbb{N}^+)$  and prove it is  
 true for  $n=k+1$ .

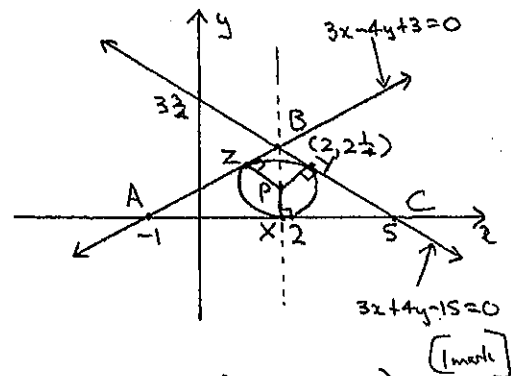
Now if  $n=k+1$   $U_n = U_{k+1}$   
 $= 5U_k - 6U_{k-1}$   
 $= 5[3^k - 2^k] - 6[3^{k-1} - 2^{k-1}]$   
 $= 5 \cdot 3^k - 5 \cdot 2^k - 6 \cdot 3^{k-1} + 6 \cdot 2^{k-1}$   
 $= 5 \cdot 3^k - 2 \cdot 3^k$   
 $= 3 \cdot 3^k - 2 \cdot 2^k$   
 $= 3^{k+1} - 2^{k+1}$

[2 marks]

$\therefore$  if it is true for  $n=k$  so it is  
 true for  $n=k+1$ .

Step 3: It is true for  $n=1, 2$  and  $3$   
 so it is true for  $n=3+1=4$ . It is  
 true for  $n=4$  and so it is true for  
 $n=4+1=5$  and so on for all  
 positive integral values of  $n$ .  
 [1 mark]

8(c)



[1 mark]

Let  $A = (-1, 0)$ ,  $C = (5, 0)$ 

$$3x - 4y + 3 = 0 \quad \text{--- (1)}$$

$$3x + 4y - 15 = 0 \quad \text{--- (2)}$$

$$\text{①} + \text{②}: 6x - 12 = 0 \quad \therefore x = 2 \text{ sub into ②}$$

$$\therefore y = 2.5 \quad \therefore B = (2, 2.5)$$

$$\text{Now } d_{AB} = \sqrt{(2 - (-1))^2 + (2.5 - 0)^2} = 3.5$$

$$d_{BC} = \sqrt{(5 - 2)^2 + (0 - 2.5)^2} = 3.5$$

 $\therefore \triangle ABC$  is isosceles

$\therefore x=2$  is the right bisector [1 mark]  
 of side AC.

Let  $P(2, y_1)$  be the centre of  
 the inscribed circle.

Let  $X, Y$  and  $Z$  be the feet of  
 the perpendiculars drawn from  $P$  to  
 each line.

Now perp. distance  $PX = PY = PZ$ 

$$\therefore |y_1| = \frac{|6 + 4y_1 - 15|}{\sqrt{3^2 + 4^2}} = \frac{|6 - 4y_1 + 3|}{\sqrt{3^2 + 4^2}} \quad \text{[1 mark]}$$

$$\therefore |y_1| = \frac{|4y_1 - 9|}{5} = \frac{|9 - 4y_1|}{5}$$

$$\therefore 25y_1^2 = 16y_1^2 - 72y_1 + 81$$

$$\therefore 9y_1^2 + 72y_1 - 81 = 0 \quad \therefore 9(y_1 + 9)(y_1 - 1) = 0$$

$$\therefore y_1 = 1 \quad (\text{from diagram}) \quad \text{[1 mark]}$$

 $\Rightarrow$  centre of inscribed circle is  $(2, 1)$ and equation is:  $(x-2)^2 + (y-1)^2 = 1$ . [1 mark]